

MATTE SIVA SAI PRAKASH

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Career Objective

Electronics and Communication Engineering student seeking opportunities in **VLSI / Embedded / Hardware Engineering**, with a strong interest in designing and developing **reliable, real-time hardware systems** and strengthening expertise in **VLSI and embedded technologies**. .

Education

SRKR Engineering College, Bhimavaram
B.Tech in Electronics and Communication Engineering
KSN Junior College, Samarlakota
Intermediate (MPC): 923/1000

2027
CGPA: 8.35/10
2021–2023

Achievements

SMART INDIA HACKATHON (SIH) 2025 – GRAND FINALE WINNER

- Selected among **top national teams** after multi-stage evaluation
- Evaluated by **government, industry & academic experts**

Projects

SIH 2025 – Rooftop Solar Phase Balancing

IoT, Power Systems, Python, OpenDSS

- Designed an **IoT-based smart phase balancing system** for 150+ MW Kerala rooftop PV grid integration
- Achieved real-time voltage monitoring and automated load switching, reducing voltage deviation to ± 2
- Benefited distribution utilities, grid operators, and rooftop solar consumers through improved grid stability and transformer utilization.

Secure IoT-based DER Monitoring & Control System

2024

IoT, Arduino, MATLAB

- Developed a secure IoT platform for real-time monitoring and control of distributed energy resources (DERs).
- Implemented encrypted communication, role-based access control, and real-time alerts to prevent unauthorized grid operations.
- Supported utility operators and grid administrators by improving system security and operational reliability.

Adaptive Vehicle Headlight Control using Sensors

2023

Embedded Systems, Sensors

- Built an embedded system to automatically adjust vehicle headlight brightness using ambient light and proximity sensors.
- Executed PWM-based control to reduce glare and enhance night-driving visibility.
- Improved safety for drivers and pedestrians by minimizing headlight glare in real-world driving conditions.

Laser-Based Home Safety Device – CRUX Innovation Challenge 2024

Embedded Systems, Arduino

- Designed and implemented a low-cost embedded home intrusion detection system using laser and LDR sensors.
- Developed a working hardware prototype that detects intrusions and triggers instant buzzer alerts.
- Enhanced home security for homeowners and residential communities with a simple and affordable solution.

Technical Skills

Programming: C, Python, MATLAB

Embedded Systems: Arduino, Microcontrollers, Sensors, Embedded C, PWM, IoT Fundamentals

VLSI & Digital: Digital Electronics, Basic VLSI Concepts, Introductory RTL Design, Verilog (Basic)

EDA & Simulation Tools: Xilinx Vivado (Basics), Cadence (Basics), Multisim, OpenDSS

Certifications & Activities

- VLSI Beginners Certification — NIELIT
- SDP on Internet of Things
- Executive Body Member, **IETE Student Forum**;
- NSS Volunteer

SOFT SKILLS

- Communication Leadership Financial Management Time Management Adaptability